24.12.2026

EEE412 Microwave Electronics- Lab 2 Experimental Report

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"I completed this homework assignment independently."

Part 1

1.a)

Network analyzer measurement are as below, at my voltage as in preliminary work (5V selected).

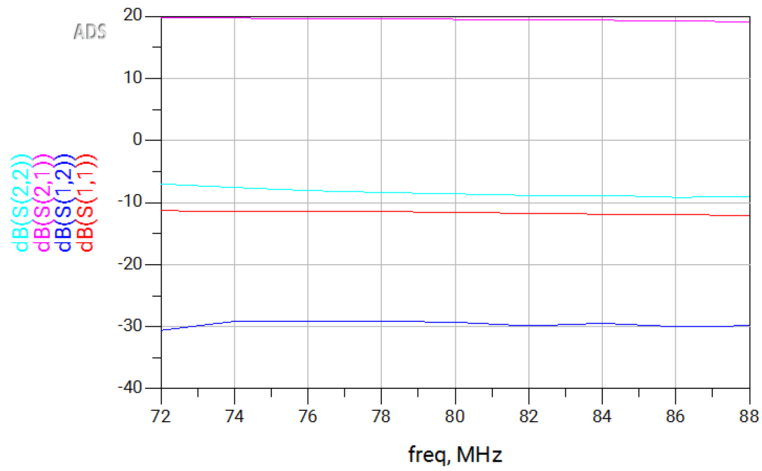


Fig.1: S parameters at Vcc= 5V

The S parameters of the BFU520-based amplifier was measured using a network analyzer at the selected center frequency of 80 MHz with a ±10% frequency span. The supply voltage was set to VCC = 5V.

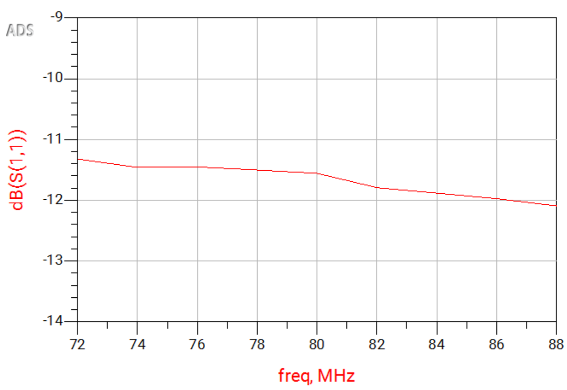


Fig.2: S11 at 5V, 80MHz

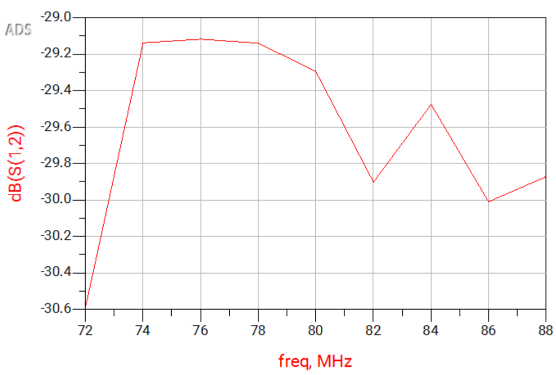


Fig.3: S12 at 5V, 80MHz

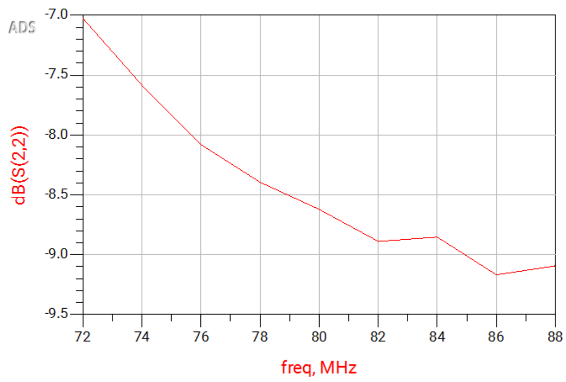


Fig.4: S22 at 5V, 80MHz

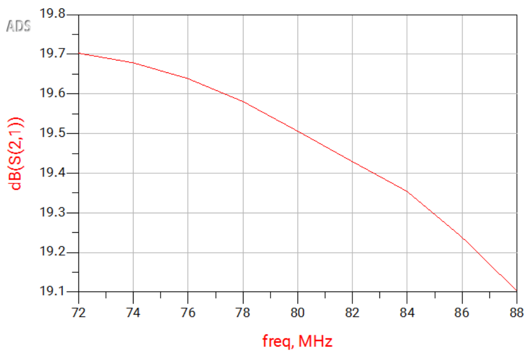


Fig.5: S21 at 5V, 80MHz

1.b) In this part, another voltage value selected and ADS measurements are as below,

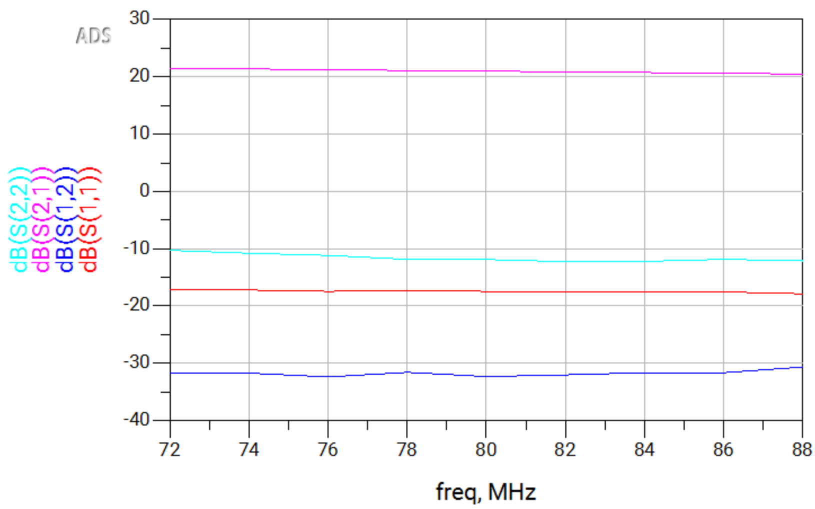


Fig.6: S parameters at Vcc= 9V

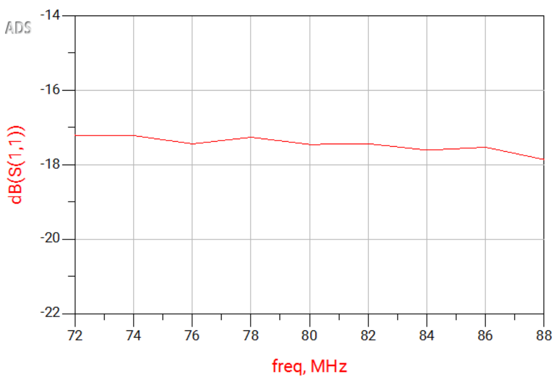


Fig.7: S11 at 5V, 80MHz

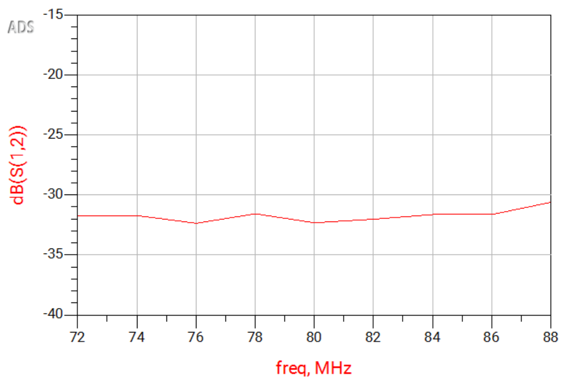
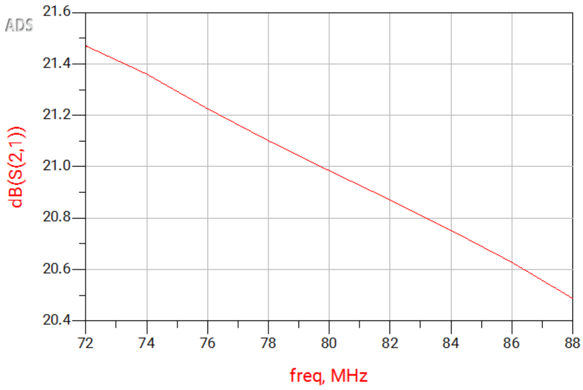


Fig.8: S12 at 5V, 80MHz



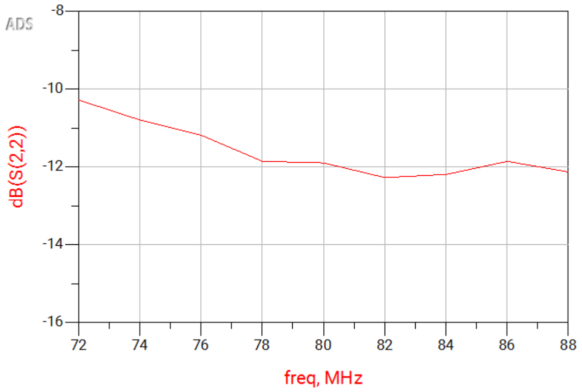


Fig.9: S21 at 9V, 80MHz

Fig.10: S12 at 9V, 80MHz

1.d) In this part 1-dB compression point measurements are made, see below for results. Starting from 5V, increasing with 1step V. Results and plots are as below. Figure 11 is the result of part c (5V).

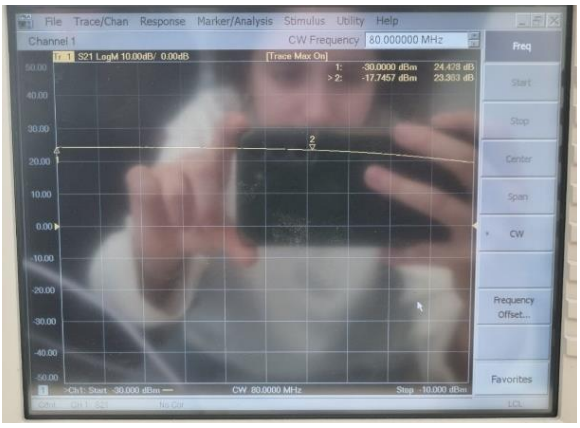


Fig.11: Vcc=5V, 80MHz

Fig.12: Vcc=6V, 80MHz

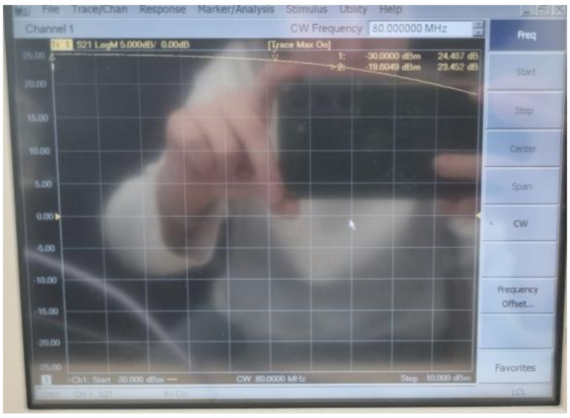


Fig.13: Vcc=7V, 80MHz

Fig.14: Vcc=8V, 80MHz

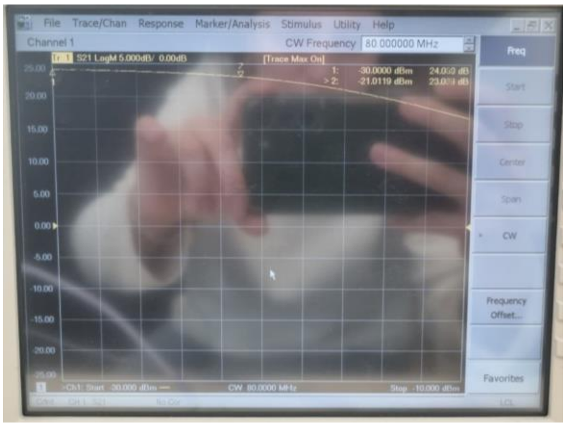


Fig.15: Vcc=9V, 80MHz

Fig.16: Vcc=10V, 80MHz

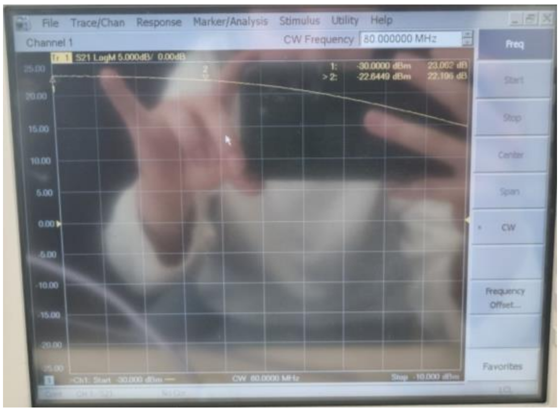


Fig.17: Vcc=11V, 80MHz

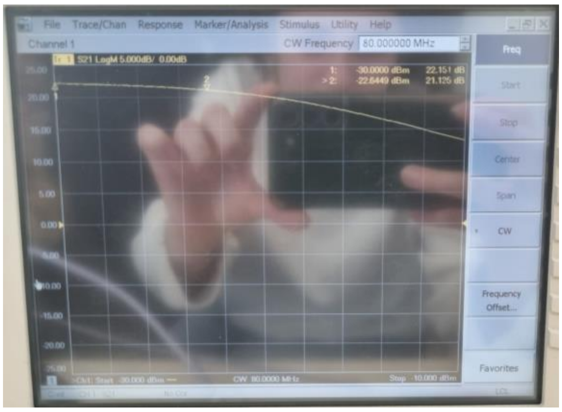


Fig.18: Vcc=12V, 80MHz

Vcc_V	Pin_1dB_dBm	Gain_dB	P1dB_Output_dBm
5	-22.645	21.125	-1.5199
6	-22.645	22.196	-0.449
7	-21.288	22.595	1.307
8	-21.012	23.039	2.027
9	-20.233	23.287	3.054
10	-19.605	23.452	3.847
11	-18.85	23.295	4.445
12	-17.746	23.363	5.617

Table 1: 1dBCompression Points From Figures Above

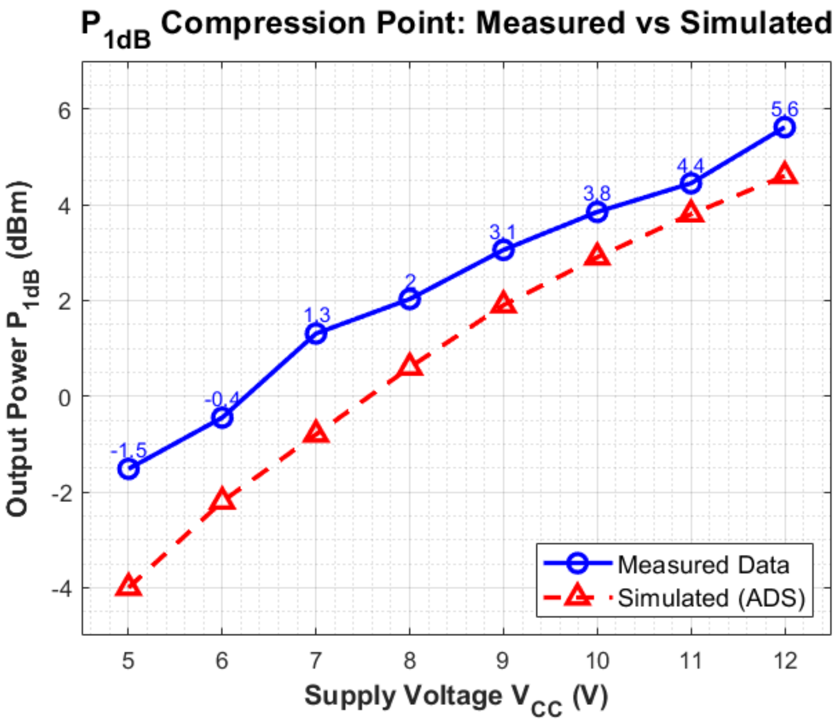


Fig.19: Simulation and Measured Value Comparison for P1dB

Part 2

2.a) Two signal generators were used to apply a two-tone input at (f_1 = 80) MHz and (f_2 = 81) MHz. In the experiment, both sources were set to an output power of 37 dBm. This level was chosen instead of 40 dBm to avoid overdriving the setup and to better capture the influence of other noise contributions.

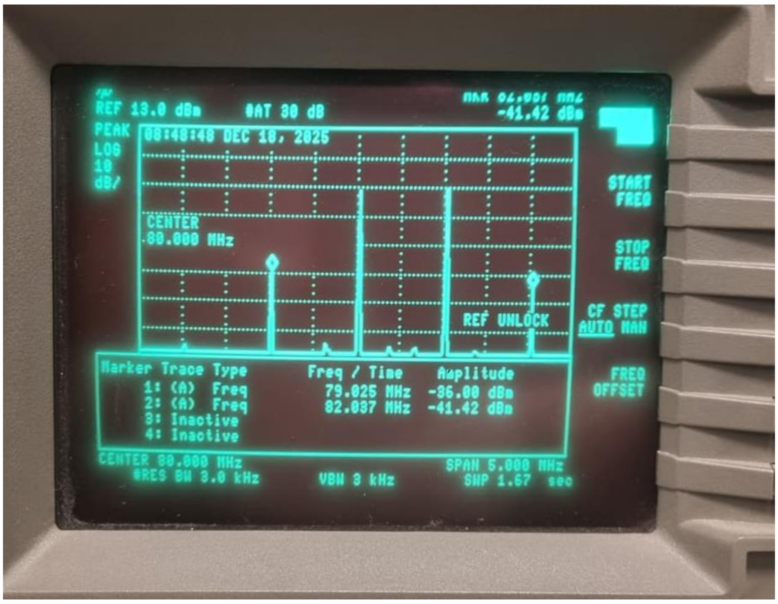


Fig.20: Fundamental and Third-order Components for -30dBm

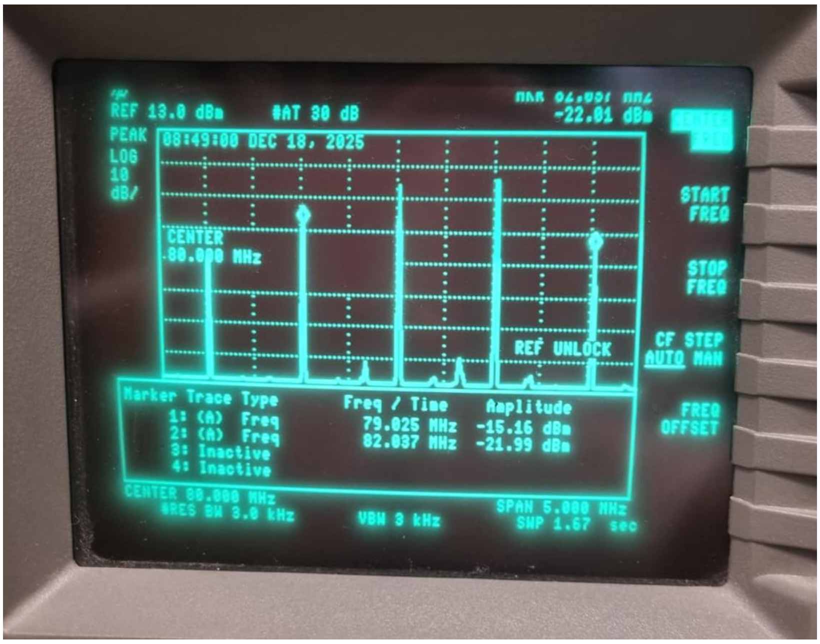


Fig.21: Fundamental and Third-order Components for -20dBm

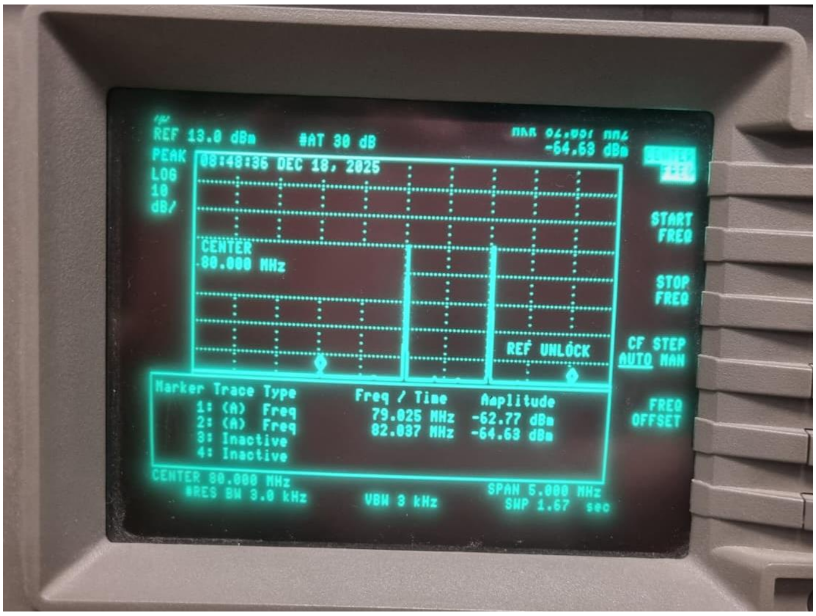


Fig.22: Fundamental and Third-order Components for -40dBm